KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING
DEPARTMENT OF AGRICULTURAL ENGINEERING

AE 451 IRRIGATION AND DRAINAGE

CLASS ASSIGNMENT 1

Problem: Soil samples were taken at different locations from a 0.4 ha farm using the core samplers (10 cm diameter x 5 cm length) about 48 hours after a rainfall event. The following measurements were made before and after oven drying the samples:

Sample No.	Mass of wet sample (gm)	Mass of dry sample (gm)
1	651.49	549.78
2	644.90	541.93
3	632.47	530.15
4	613.12	510.51
5	647.92	545.85
6	628.83	526.22
7	593.96	490.88
8	602.19	502.66
9	572.56	471.24
10	644.44	542.00

a) Calculate the average depth (mm) and total volume (m^3) of water present in 50 cm depth of the soil in the farm at the time of taking the soil samples. Present your answer in a TABULAR form ONLY.
b) Assuming the soil water conditions at the time of sampling represent the field capacity water content of the soil and assuming further that the average permanent wilting point water content of the soil on the farm is 8.73 % by mass, compute the average water content of the soil at irrigation for an allowable soil water depletion factor and average rooting depth of 0.55 and 1.2 m respectively.
c) Calculate the average net irrigation requirement (mm) on the farm.
d) Calculate the gross amount of irrigation required (mm) if irrigation is applied at 70 % efficiency.

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CLASS ASSIGNMENT 2

1. Design a *lined rectangular* canal to carry 0.93 m^3/s of water for the following conditions: Flow depth = 0.5 m; Canal bed slope = 1:500; Manning's roughness coefficient = 0.015. Assume a freeboard of 20% and check for the *maximum* permissible velocity criterion.

$$Q = \frac{A}{n} R^{2/3} S^{1/2}$$

2. You are required to design an *unlined trapezoidal* irrigation canal to be excavated in silty sand using the following design data:

Critical tractive force on channel bed	=	2.4 N/m^2
Angle of internal soil friction	=	30°
Canal discharge	=	10 m^3/s
Canal bed slope	=	1/10,000
Manning's roughness coefficient	=	0.02
Side slope of canal	=	½
Freeboard	=	20 % of design depth

Also check that the flow velocity is *non-erodible* if the *maximum non-erodible* velocity for the soil is 1.0 m/s.

$$\tau_{cs} = \tau_{cb} \sqrt{1 - \frac{\sin^2 \theta}{\sin^2 \varphi}}$$

$$\tau_{cs} = 0.76 \rho g d S$$

For the same conditions stated above, design also an *economic* channel section using values of canal discharge, canal bed slope, Manning's roughness coefficient, side slope and freeboard stated above and compare the two results.

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CLASS ASSIGNMENT 3

Problem: Infiltration data obtained from a field measurement using the double ring infiltrometer is provided in the table below.

SN	Time (t, min)	Infiltration rate (i, mm/min)
1	1	3.80
2	2	2.39
3	3	1.68
4	4	1.63
5	9	1.38
6	14	1.18
7	19	1.10
8	29	0.98
9	39	0.88
10	49	0.84
11	60	0.80
12	80	0.68
13	100	0.67
14	120	0.63

1. Using the Microsoft Excel Software, plot the following curve:
    - infiltration rate (*i*, mm/min) against time (*t*, min)
2. Fit the curve with an appropriate trend line, generating the *Kostiakov* equation and its r^2 value.
3. Determine the infiltration rate after an elapsed time of 3 hours.
4. Derive the cumulative infiltration (*d*, mm) versus time (*t*, min) equation from the equation obtained in step 2 above.
5. How long will it take for 75 mm of irrigation water to be infiltrated into this soil (h)?

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